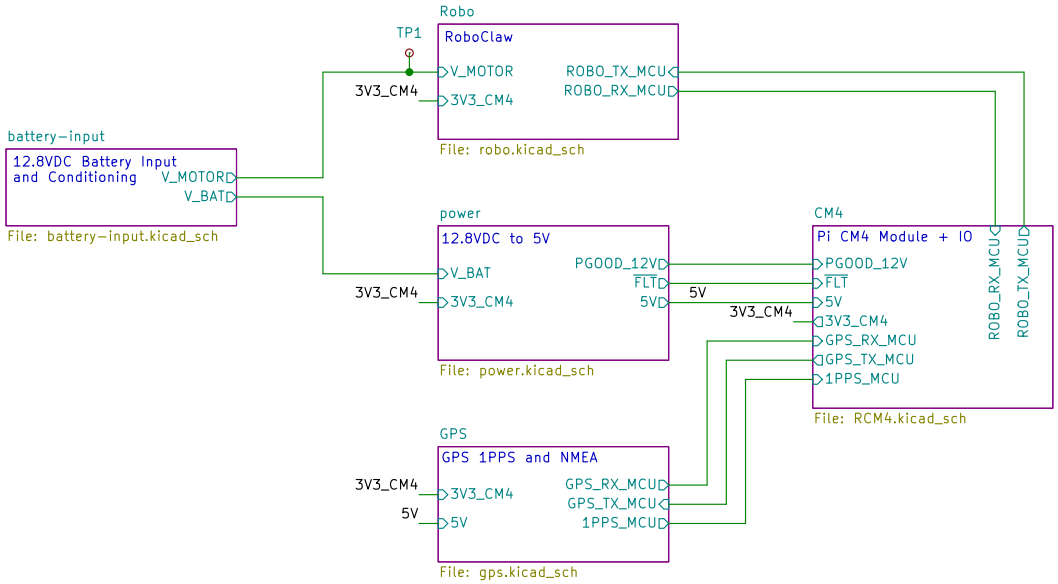
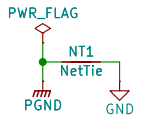
Raspberry Pi CM4 + RoboClaw Carrier Board (ROBOCLAW-2X15A-OEM)



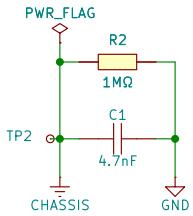
Motor Notes

File: motor-notes.kicad_sch

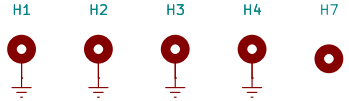
Grounding



NT1: single-point PGND-GND tie. Place at J1-2 (battery -). Motor current (RoboClaw BAT-) stays on PGND.



R2/C1: CHASSIS-GND static bleed (R2) and HF bond (C1, keep 2 kV). Shields and RJ45 termination (P10) go to CHASSIS. Installed, GND is also bonded to the box at the J2 arrestor, so R2 is bypassed at DC.



H1/H2/H3/H4: M3 plated mounting holes on CHASSIS. Bond to the metal enclosure or dish frame.
H7: M2.5 hole, no copper, for the standoff of a J4 HAT board (M2.5 x 11 mm).

Sheet: / File: cm4-robo-controller.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/13

CM4

Suggest a CM4004008.
4GB RAM, 8GB eMMC program storage.

A2A
ComputeModule4-CM4
Hirose
2x DF40C-100DS-0.4V

A2B
ComputeModule4-CM4
Hirose
2x DF40C-100DS-0.4V

GPIO

High Speed Serial

NB SD signals are only available on modules without eMMC

J17
CM4 receptacle
BOM / CPL only
DF40C-100DS-0.4V(51)

J18
CM4 receptacle
BOM / CPL only
DF40C-100DS-0.4V(51)

Module1 pins 89 (WiFi_nDisable) and 91 (BT_nDisable) are left unconnected on purpose. This board needs a CM4 WITHOUT wireless, where both pins are reserved (CM4 datasheet 2.1.1 / 2.1.2). If a wireless CM4 is ever fitted, disable the radios in config.txt: dtoverlay=disable-wifi and dtoverlay=disable-bt (the latter also frees UART0).

J17/J18: Hirose DF40C-100DS-0.4V(51) CM4 receptacles (LCSC C597931). BOM / CPL only: their pads are in the A2 footprint. J17 = pads 1-100, J18 = pads 101-200, pin 1 lower left. A1 (RoboClaw) and A2 (CM4): DNP, bought separately, fitted by hand: A2 is also excluded from the position file.

CM4 power-health LED (PI_LED_nPWR, buffered per CM4 datasheet)

From the CM4 Manual: Active-low output to drive Power On LED. This signal needs to be buffered.

74LVC1G07SE-7

File: usb.kicad_sch

File: ethernet.kicad_sch

File: CM4_Connectors

File: i2c.kicad_sch

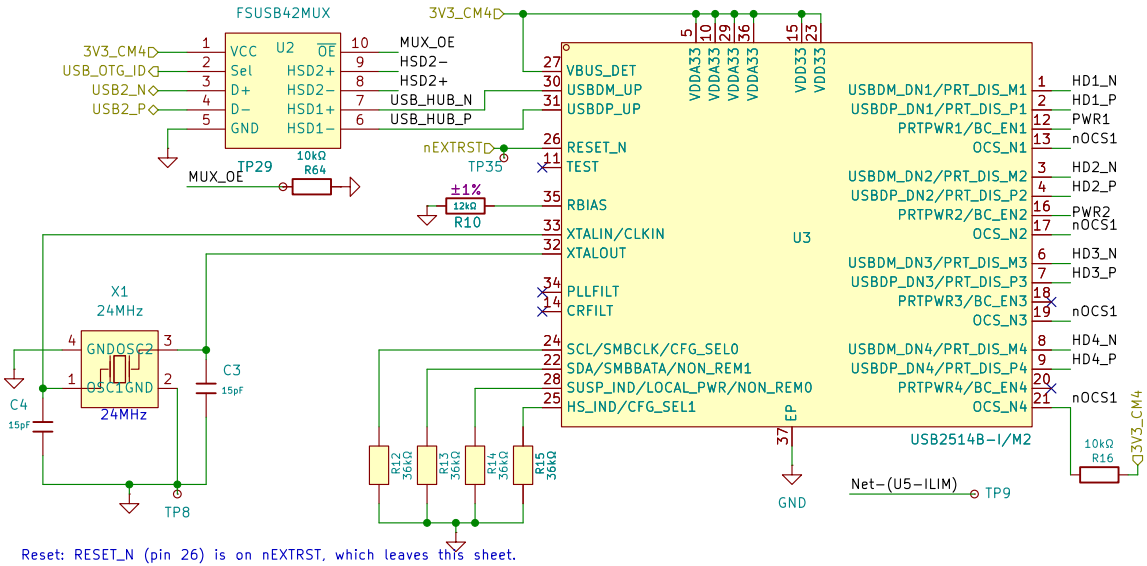
Sheet: /CM4/
File: RCM4.kicad_sch

Title:

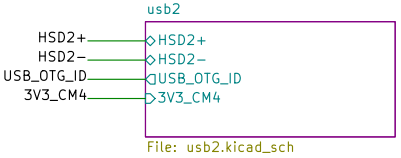
Size: A4 | Date: | Rev:
KiCad E.D.A. 10.0.1 | Id: 2/13

USB — 2 Ports

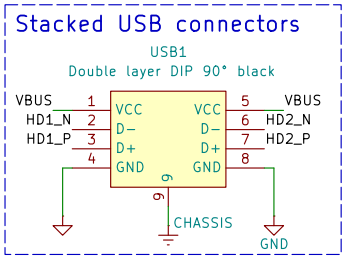
4-Port USB Hub (all 4 ports used)

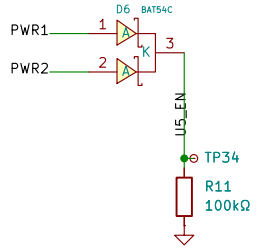


Reset: RESET_N (pin 26) is on nEXTRST, which leaves this sheet.



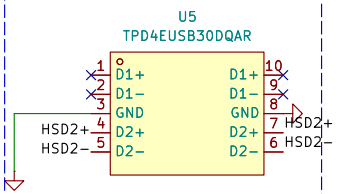
Stacked USB connectors



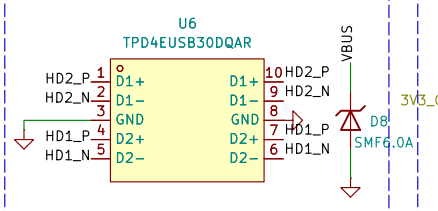


U4 port power, diode-OR (D6, R11). The hub straps CFG_SEL=00: individual port power switching and OC sensing (USB2514B Table 5-1). U4 feeds both USB1 ports, so its EN is the OR of PRTPWR1-2: VBUS stays on while either port is enabled and drops only when both are off. Ports 3 and 4 are disabled by strap, so PRTPWR3/4 are left open. OCS_N1-4 share nOCS1: a U4 fault is reported on every port (correct for one shared switch). BC_EN straps read 0 via the internal 10k IPD.

USB-C ESD



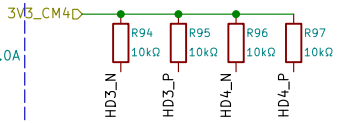
USB-A ESD



Protects both USB-A ports (4 lines).
Diode clamps VBUS. Place both at USB1.

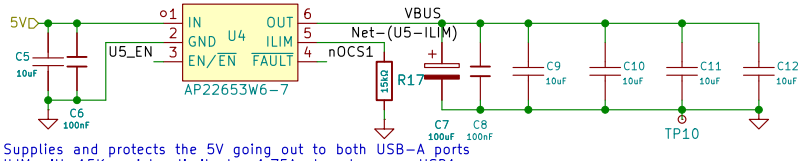
Place ESD protection next to respective USB ports

Unused hub ports 3 and 4



Hub ports 3 and 4 are not used. 10k pull-ups on D+ and D- disable them: the PRT_DIS straps are sampled at RESET_N (USB2514B, CFG_SEL=00).

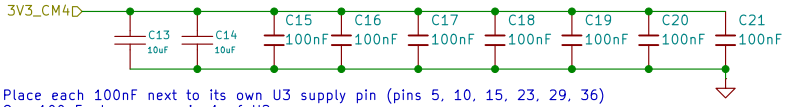
Current-limited high-side switch



Supplies and protects the 5V going out to both USB-A ports
ILIM with 15K resistor limits to ~1.75A shared across USB1

C7 SUBSTITUTION (2026-11-01)
Preferred part: Panasonic EEFCX0J101R (LCSC C189619), 100uF 6.3V polymer, 15mOhm, 5.1A ripple. Out of stock at JLCPCB since 2026-10-30 (min order 14), so AISHI SA11AM101A19R15XXX (C135491) is fitted: same 7.3x4.3mm D-case footprint, 100uF 10V, 15mOhm, 2.5A ripple, which is ample for VBUS hold-up. Refit the Panasonic when it is back in stock; nothing else changes.

3.3V Caps for U2 and U3 VDD/VCC.



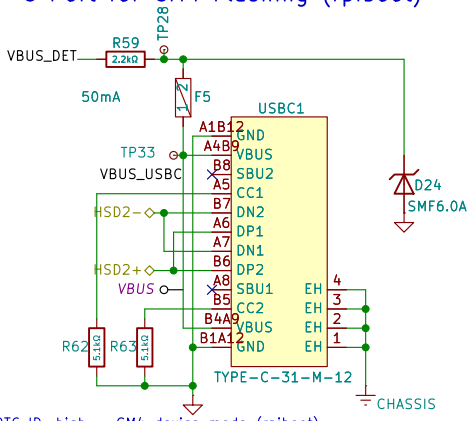
Place each 100nF next to its own U3 supply pin (pins 5, 10, 15, 23, 29, 36)
One 100nF place near pin 1 of U2

Sheet: /CM4/usb/
File: usb.kicad_sch

Title: USB

Size: A4	Date: 2026-09-20	**Rev: 1.1**
KiCad E.D.A. 10.0.1		Id: 3/13

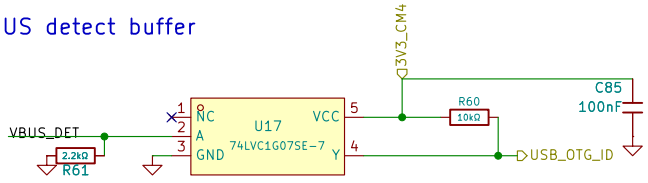
USB-C Port for CM4 Flashing (rpiboot)



USB_OTG_ID: high = CM4 device mode (rpiboot), low/GND = CM4 host. Needs otg_mode=1 in config.txt.
USBC1 VBUS: 5.0 V nom (4.75 V min / 5.5 V max).
No USB-PD controller; CC has 5.1 kΩ Rd only.
VBUS detect: R59/R61 (2.75 V max) -> U17 open-drain -> USB_OTG_ID.
VBUS current: 1.25 mA continuous / 1.25 mA peak.
VBUS survives 6.0 V DC (D24 SMF6.0A stand-off), surges clamped to 10.3 V; -0.7 V (D24 forward). U17 A sees half via R59/R61.

F5 (TECHFUSE SMD0603-005-24V, 50 mA hold, 200 mA trip, 2-4 ohm) sits between USBC1 VBUS and the D24 / R59 / TP28 node. A short on that node (D24 failed, TP28 probe slip) trips F5 instead of relying on the USB source current limit. At 1.25 mA F5 drops < 10 mV, so VBUS detect is unaffected; its 24 V rating covers the 6 V D24 stand-off. Nothing on the board draws power from VBUS. VBUS_USBC is the connector-side net.

VBUS detect buffer



U17 is open-drain, powered from 3V3_CM4. With the CM4 off, Y is high-Z and R60 is unpowered, so U2 Sel and CM4 USB_OTG_ID are never driven (CM4 datasheet 5.1). 74LVC Ioff: A tolerates VBUS_DET at VCC = 0.

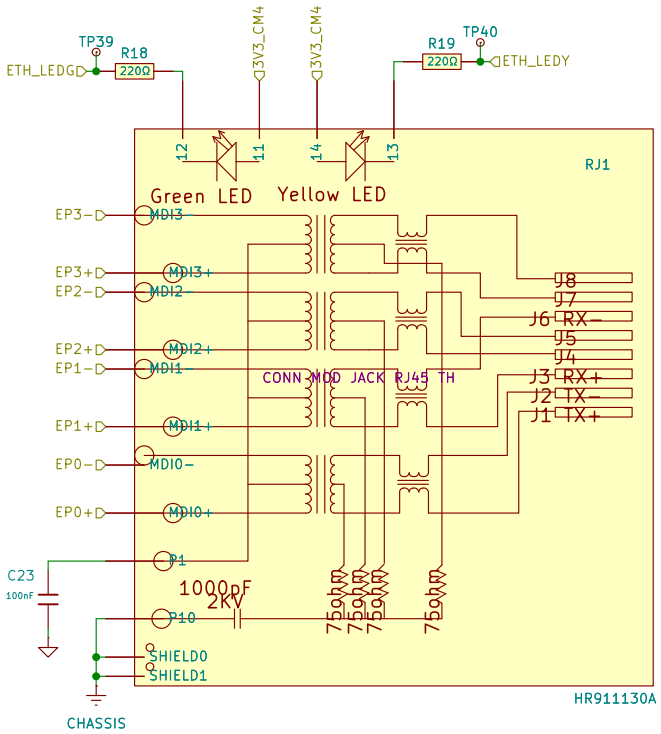
Sheet: /CM4/usb/usb2/
File: usb2.kicad_sch

Title: USB-C rpiboot port

Size: A4 | Date: 2026-10-13 | **Rev: 1.1**
KiCad E.D.A. 10.0.1 | Id: 10/13

Gigabit Ethernet — CM4 MDI to RJ45

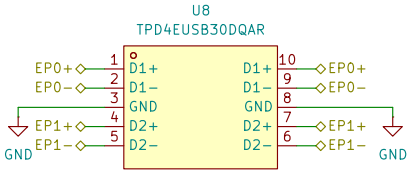
RJ45 with integrated magnetics

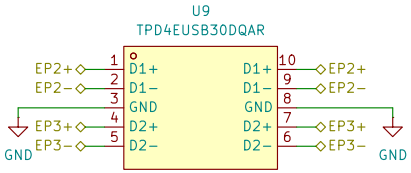


HR911130A, shielded, integrated magnetics and LEDs, 100/1000Base-T (LCSC C54408).
EP0–EP3 come straight from the CM4 – the PHY is on the module, no external PHY.
Chip-side center taps (P1) bypassed to GND by C23 100nF.
Bob Smith termination (4x75R + 1000pF/2kV) is built into the jack;
P10 is that node, tied to CHASSIS. Shield pins are also on CHASSIS
(bled to GND by R2/C1 on the top sheet).

CM4 Ethernet_nLED pins are active-low 3.3V sinks, IOL 8mA @ VOL 0.4V.
ETH_LEDG = activity (green, RJ1 pin 11 anode / 12 cathode).
ETH_LEDY = link speed (yellow, pin 14 anode / 13 cathode).
R18/R19 220R gives ~4.5mA through the LED. That is below the 8mA limit.
Route EP0–EP3 as 100R differential pairs, matched within 0.15mm.
Keep the RJ1 chassis/shield area clear of the signal ground pour.

Ethernet ESD





Clamps all four pairs on the CM4 side of the magnetics.
Place both at RJ1.

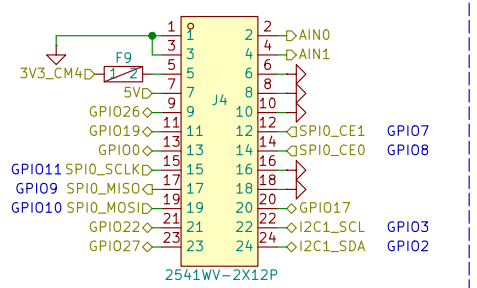
Matches CM4 datasheet Fig. 2 (TPD4EUSB30 on all pairs).
TPD4EUSB30 IO abs max is 0 to 6 V (TI p.4), so EP lines must stay
>= 0 V. MDI DC level is set inside the BCM54210PE
(not published; CT is AC-grounded only).
First boards: scope each EP line to GND at the clamps
at 1000/100/10 (ethtool -s eth0 speed N duplex full
autoneg off). Minimum must be >= 0 V (no undershoot below GND).

Use JLC's impedance calculator for the stackup to get
width and gap for 100 Ω (Ethernet) and 90 Ω (USB) differential
on the outer layers. Put those values into ETH_100_OHM and USB_90_OHM.
Order with impedance control enabled, so JLC adjusts to the target.

Sheet: /CM4/ethernet/ File: ethernet.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 4/13

CM4 External Header Pins

GPIO Headers



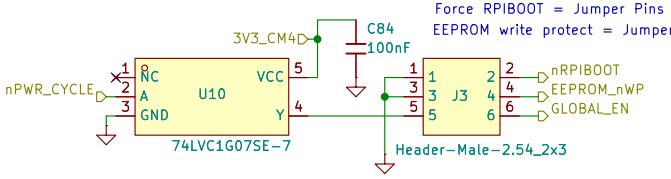
F9 (100 mA PTC, SMD0603-010-24V) fuses the 3.3 V on J4 pin 5: a short on a plug-in board trips F9, not the CM4 rail. HAT budget 100 mA; CN3 has F8.

AIN0 / AIN1 (J4-2 / J4-4), field wired, to CM4 AnalogIP0/1 (MxL7704 ADC). Measures 0 to 2.55 V (8-bit). Survives +/-24 V either polarity: R7/R6 10k series + D4/D5 BAT54S clamps to GND and 3V3_CM4 (RCM4 sheet) hold the pins within −0.3 V / +3.6 V (abs max −0.3/+6.0). Error: ADC input < 1 uA x 10k = 10 mV, plus clamp leakage (up to -1-2 % hot). With the CM4 off, a driven input feeds < 0.3 mA (at 6 V) into 3V3_CM4 through the upper clamp.

GPIO0 (J4-13) as plain GPIO: add to config.txt
force_eeprom_read=0
disable_poe_fan=1
or the firmware probes GPIO0/1 (I2C0) for a HAT EEPROM at boot.

Boot/EEPROM_EN/GLOBAL_EN

Force RPIBOOT = Jumper Pins 1-2
EEPROM write protect = Jumper Pins 3-4



POWER-CYCLE CONTROL (nPWR_CYCLE)
Lets the CM4 cut its own power after a clean shutdown and restart by itself. Use it when a normal reboot is not enough: it fully resets the PMIC and all on-module power rails.

How it works:
- GPIO6 drives U10 (74LVC1G07, open-drain). J3 pins 5-6 connect U10's output to GLOBAL_EN (CM4 pin 99, 100k internal pull-up to 5V).
- GPIO6 high (default pull-up at reset): U10 output is off and the CM4 runs.
- GPIO6 low: GLOBAL_EN is pulled low and the CM4 powers off.
- U10 is powered from 3V3_CM4, which switches off with the CM4, so U10 releases GLOBAL_EN and the CM4 powers back on.

Use:
- Fit the J3 5-6 jumper to enable. Fit J3 5-6 only after the board and GPIO6 behaviour have been verified.
- config.txt: dtoverlay=gpio-poweroff,gpiopin=6,active_low=1 "sudo poweroff" then shuts down cleanly and power-cycles the CM4.
- Do not drive GPIO6 low while the OS is running. CM4 datasheet section 2.13: pull GLOBAL_EN low only after the OS has shut down.
- The datasheet needs GLOBAL_EN low for >1 ms. The low time here is set by how fast 3V3_CM4 decays and is not guaranteed: check it with a scope on first boards.

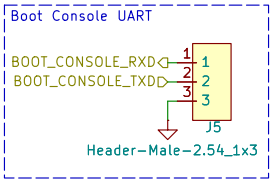
EEPROM WRITE PROTECT (EEPROM_nWP, J3 3-4)
The CM4 pulls EEPROM_nWP up internally (100k), so the bootloader EEPROM is writable unless J3 3-4 is fitted. The jumper only takes effect once protection is enabled in the bootloader config (CM4 datasheet 2.18).

Production:
- Set eeprom_write_protect=1 in the bootloader config (via usbboot or rpi-eeprom-config) before shipping.
- Fit the J3 3-4 jumper. The EEPROM is then protected from power-up (CM4 datasheet 5.1).

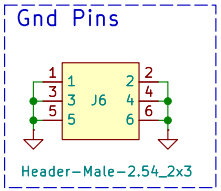
Bootloader update:
- Remove J3 3-4, update via usbboot or self-update, then refit the jumper.

Why not a GPIO with a pull-down: CM4 EEPROM updates run only via usbboot or bootloader self-update (datasheet 6.3), before Linux can drive a GPIO, so the pin could never be released for an update.

Boot Console UART



Gnd Pins



I2C PULL-UPS

I2C1: GPIO2 = I2C1_SDA (J4-24), GPIO3 = I2C1_SCL (J4-22) Pulled up to 3.3 V (GPIO_VREF) by 1.8 kΩ inside the CM4 (CM4 datasheet section 2.5, pins 56/58). No board pull-ups are needed. R20/R21 (2.2k, fitted via H5/H6) are optional.

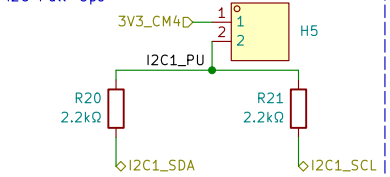
BOOT CONSOLE UART (J5): CM4 UART0, 3.3 V logic, 115200 baud, 8N1, no flow control
J5-1 BOOT_CONSOLE_RXD (GPIO15, CM4 receives) <- adapter TXD
J5-2 BOOT_CONSOLE_TXD (GPIO14, CM4 transmits) -> adapter RXD
J5-3 GND -- adapter GND
TX and RX cross over: the adapter TXD goes to the CM4 RXD.

Use an inexpensive USB-to-TTL serial cable (PL2303 / CP2102 / FTDI type) with female jumper leads. Typical colours: black GND, white RXD, green TXD, red 5V. Check the labels on your cable.

DO NOT connect the adapter 5V lead. The board powers itself and J5 has no 5V pin. A loose 5V lead can short to the header or the board, so it is best to cut it off (or insulate it).
Use an adapter with 3.3 V signals only. CM4 GPIOs are not 5 V tolerant: check that the adapter TXD idles at about 3.3 V with a meter before connecting.

Linux console: enable_uart=1 in config.txt (Pi OS cmdline.txt already has console=serial0,115200).
Bootloader messages: set BOOT_UART=1 in the EEPROM config (rpi-eeprom-config).

I2C Pull-Ups



H5: optional I2C1 pull-ups R20/R21, normally OPEN. CM4 already pulls GPIO2/3 up with 1.8k (DS p.7). Fit H5 only for a long, high-capacitance bus at 400 kHz: -1k total, -2.9 mA sink at 0.4 V (I2C spec min 3 mA, so compliant devices are OK).

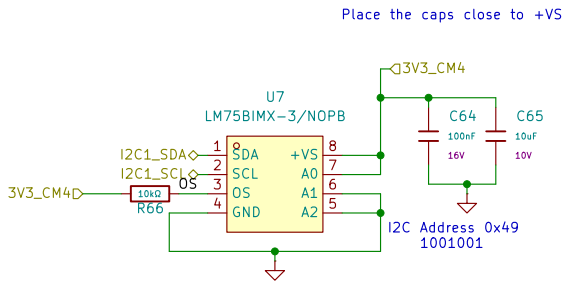
Sheet: /CM4/CM4_Connectors/
File: cm4_connectors.kicad_sch

Title:

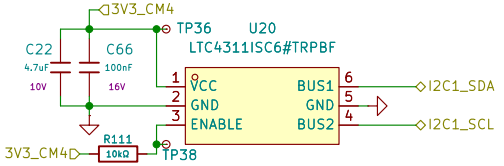
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 5/13

I2C

On-Board Temperature Sensor



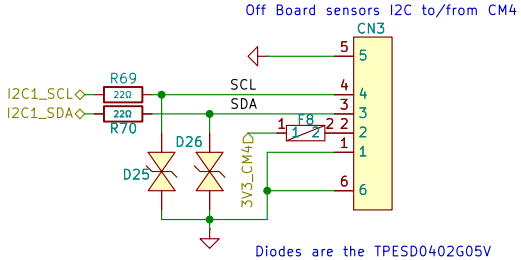
I2C Extender



U20 ENABLE: R111 10k pull-up to 3V3_CM4, TP38 for test (the unlabelled pad right of R111, just below U20). Ground TP38 to shut the LTC4311 down (BUS pins high-Z); release to run.

I2C1 pull-ups: the CM4 pulls GPIO2/3 up with 1.8 kΩ. H5 can add R20/R21 (2.2 kΩ, normally open) on the CM4 Connectors sheet. There are no pull-ups on this sheet. Off-board sensors (e.g. Adafruit breakouts) often carry their own; keep the parallel total above ~1 kΩ so a 3 mA sink still pulls the bus to 0.4 V.

I2C Off-board Connector



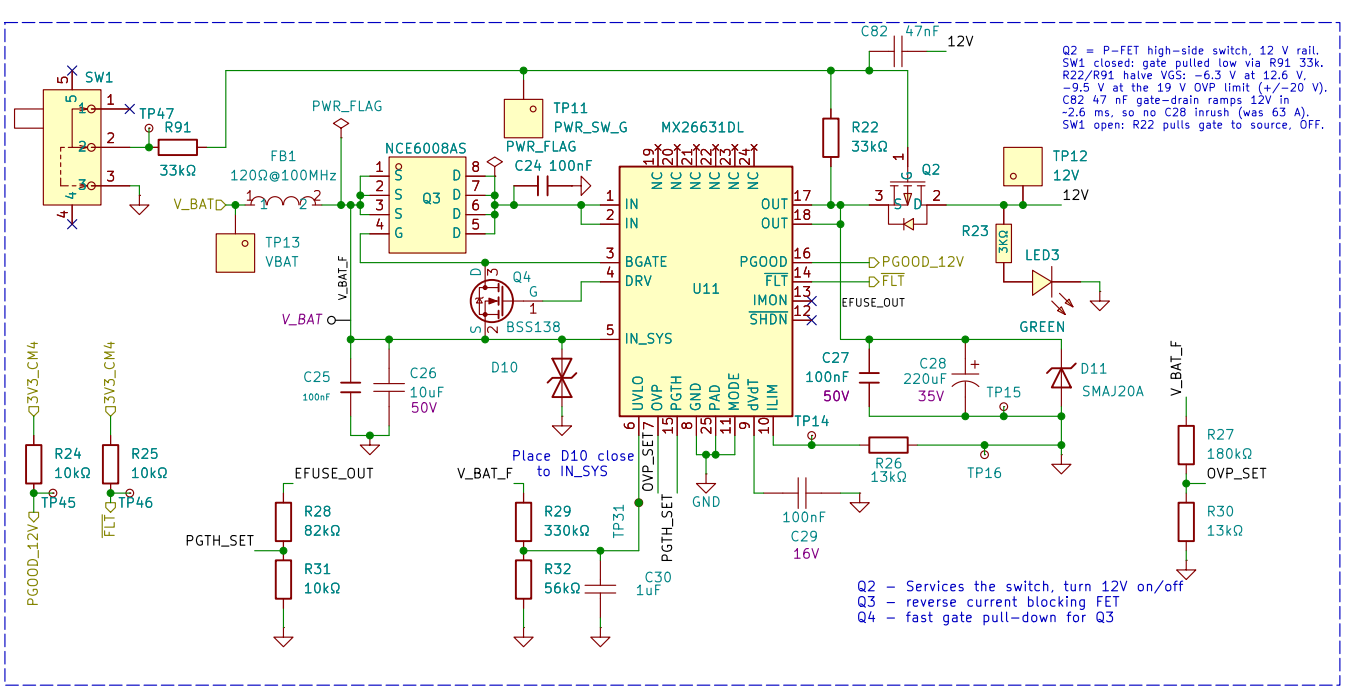
Diodes are the TPESD0402G05V

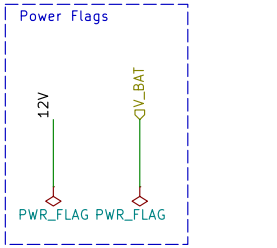
The 22Ω resistors limit fault current and damp cable reflections without meaningfully affecting I2C at the operating speed.
F8 (TECHFUSE SMD0603-010-24V, 100 mA hold, 300 mA trip, 0.9 ohm) feeds CN3 pin 2 from 3V3_CM4: a shorted or miswired sensor cable trips F8 instead of collapsing the CM4 3.3 V rail and rebooting the CM4. Sensor budget 50 mA (about 50 mV drop); the CM4 PMIC is the backstop.

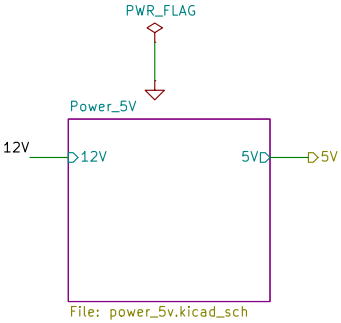
I2C pull-ups R20/R21 are on the CM4 Connectors sheet, enabled by H5.

Sheet: /CM4/I2C/		
File: i2c.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 12/13

BATT 12V In to 12V
Current limiting, On/OFF switch, Reverse polarity protection







This circuit takes 12V from the battery, VBAT.
Reverse battery is blocked by Q1 at J1 (top sheet).
The eFuse limits to 3.08A typ (2.7–3.4A, see below).

EFUSE CURRENT LIMIT: R26 13k -> IOL = 40/13k = 3.08 A typ. Est. 2.7–3.4 A, from the datasheet IOL limits at 20k (–12/+8%) and 7.5k (–10/+11%) plus R26 1%. Branch demand ~1.9 A at 10 V: >= 40% margin at the low corner.

REVERSE BATTERY: blocked by Q1 (SP40P04NK P-FET) at J1 on the top sheet, ahead of the F2/F1 split. Reversed: VGS = 0 and the body diode is reverse-biased, so no current flows, no fuse blows, and V_BAT/V_BAT_F never go negative. U11 IN_SYS/UVLO/OVP/BGATE (abs min –0.3 V) stay within rating. D2 (15 V) clamps Q1 VGS during TVS clamping (~29 V). Q1 conducts both ways when on, so RoboClaw regen still reaches the battery. Q3 keeps its reverse-current-blocking role on this branch.
F2/F1 are Keystone 3557-2 holders (ATO fuses fitted at final assembly).

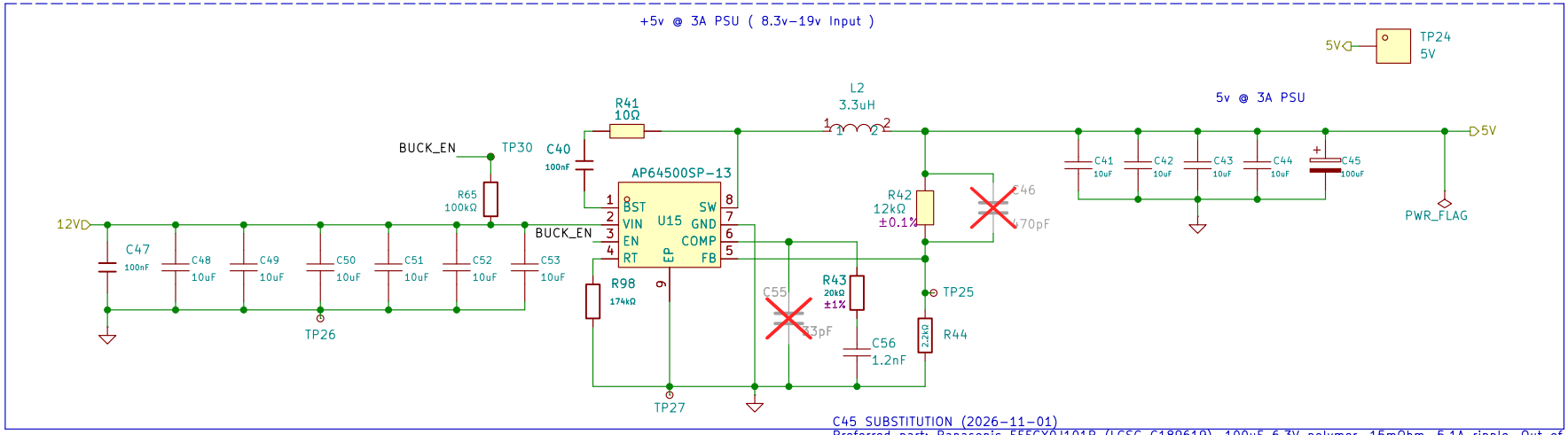
SW1 SWITCHES THE CM4 / LOGIC SUPPLY ONLY.
With SW1 off the board still draws about 31 mA from the battery: RoboClaw logic ~30 mA (fed from V_MOTOR), U11 + UVLO/OVP/PGOOD dividers ~1.3 mA. A full 12 Ah battery runs flat in about 16 days. For storage or transport, unplug J1 (or pull F1 and F2), and charge the battery first.

JRSeti
Sheet: /power/
File: power.kicad_sch

Title: Power

Size: A4	Date: 2026-08-18	**Rev: 2.0**
KiCad E.D.A. 10.0.1		Id: 6/13

12V to 5V



C45 SUBSTITUTION (2026–11–01)
Preferred part: Panasonic EEFCX0J101R (LCSC C189619), 100uF 6.3V polymer, 15mOhm, 5.1A ripple. Out of stock at JLCPCB since 2026–10–30 (min order 14), so AISHI SA11AM101A19R15XXX (C135491) is fitted: same 7.3x4.3mm D–case footprint, 100uF 10V, 15mOhm, 2.5A ripple, which is ample for the 5V buck output ripple. Refit the Panasonic when it is back in stock; nothing else changes.

12V to 5V Buck – AP64500 (5A, 575kHz set by R98, 0.8V reference)
Vout = 5.16V, designed for 3A. Input 8.3V–19V (eFuse UVLO cuts in at 8.3V, OVP cuts out at 17.5–19V).
EN pulled up to VIN by R65 (100k): starts automatically, the eFuse upstream provides
the UVLO. TP30 on EN lets a test fixture hold the buck off.

Output voltage set by the FB divider

Vout = 0.8V x (1 + R42/R44) = 0.8 x (1 + 12k/2.2k) = 5.16V
Corners 5.10–5.22V (Vref 792–808mV, R42/R44 0.1%). Worst–case USB VBUS
= 5.10V – 40mV load reg – 1.908A x 135mΩ (U4) = 4.81V >= 4.75V;
max 5.22V stays within the CM4 5.25V limit. 1% resistors cannot meet both.
R42 12k 0.1% thin film top leg, 5V to FB
R44 2.2k 0.1% thin film bottom leg, FB to GND
C46 DNP 470pF optional feedforward across R42 for transient tuning
(datasheet example 47pF across 115k, scaled for R42 = 12k)

Loop compensation

R43 20k + C56 1.2nF series RC on COMP sets the loop response
C55 DNP 33pF optional HF pole (datasheet Table 1, 5V), fit only if needed

Gate drive and soft–start

C40 100nF bootstrap cap, BST to SW – powers the high–side FET driver
R41 10R boot resistor, slows the SW rising edge to cut EMI
R98 174k RT/CLK to GND: fsw = 100000/174k = 575kHz (AP64500 Eq. 7);
never leave RT/CLK open. Soft–start is fixed at 4ms inside the AP64500

Input bulk and bypass

C48–C53 6 x 10uF 35V bulk; carries ~1.5A RMS ripple at 3A out
C47 0.1uF HF bypass, place closest to VIN/GND

Output filter

L2 3.3uH SRP7028A–3R3M Bourns molded, 6A Irms, 12A Isat, 30mΩ DCR
with 575kHz gives 1.55A p–p ripple at 12V in
(2.0A p–p at 19V in; peak coil current 4.0A at 3A load)
Isat 12A > U15 peak current limit (9.2A max), so L2
does not saturate during overload before hiccup
C41–C44 4 x 10uF 10V ceramic, low ESR
C45 100uF 10V polymer (AISHI, see note) bulk for load steps; 5.16V = 52% of rating

U15 SUBSTITUTION (2026–10–27)
Preferred part: AP64501SP–13 (LCSC C2071517), fixed 570kHz, adjustable soft–start.
It is out of stock (JLCPCB 1 pc, Mouser 0, DigiKey 0), so AP64500SP–13 (C2070920) is fitted.
Same die, pinout and SO–8EP footprint; only pin 4 differs: SS on the AP64501, RT/CLK on the
AP64500. Loop gm, current sense, Vref, limits and compensation are identical, so R42/R44,
R43/C56 and L2 are unchanged.
R98 174k sets 575kHz; spread–spectrum stays on. The external–clock sync is not used
(it needs a >3.5V clock and turns spread–spectrum off).
To go back to the AP64501: fit AP64501SP–13 and replace R98 with a 100nF 0402 cap
(CL05B104KO5NNNC, C1525) for the ~20ms soft–start.

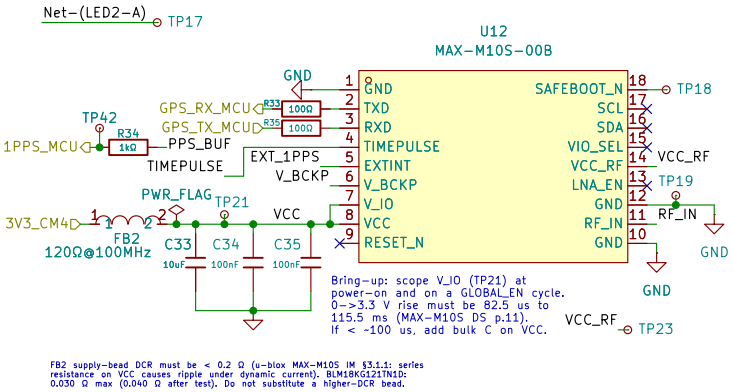
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File: power_5v.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.1 | Id: 8/13

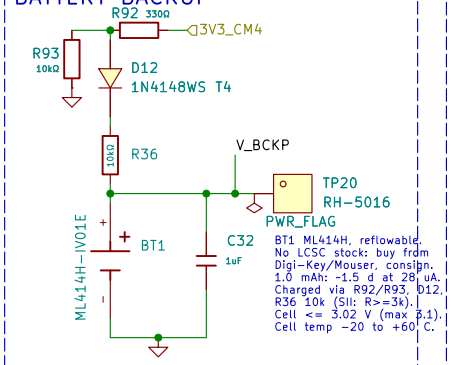
GPS

GPS is input using a MAX-M10S-00B.
NMEA sentences are used to get the UTC datetime and
the 1PPS is used to correct the MCU clock every second.
See https://www.mouser.com/pdfDocs/MAX-M10S_IntegrationManual_UBX-20053088.pdf



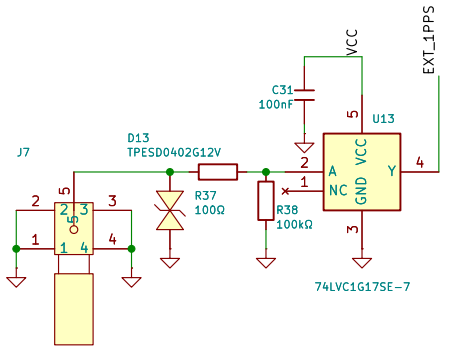
FB2 supply-bead DCR must be < 0.2 Ω (u-blox MAX-M10S IM §3.1.1: series resistance on VCC causes ripple under dynamic current). BLM18KG121TN1D: 0.030 Ω max (0.040 Ω after test). Do not substitute a higher-DCR bead.

BATTERY BACKUP



BT1 ML414H, reflowable. No LCSC stock: buy from Digi-Key/Mouser, consign. 1.0 mAh: ~1.5 d at 28 uA. Charged via R92/R93, D12, R36 10k (SII: R>=3k). Cell <= 3.02 V (max 3.1). Cell temp −20 to +60 C.

1PPS EXT INPUT (FE-5680A)

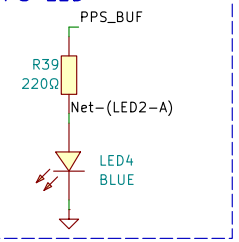


1PPS from FEI FE-5680A rubidium clock, 3-5V logic pulse. U13 Schmitt buffer (5V-tolerant input, VT+ <= 2.0V at 3V) shifts it to the 3.3V V_IO level; EXTINT abs max is V_IO+0.3V (3.6V). R38 holds the input low when nothing is connected. Edges on EXTINT are timestamped in UBX-TIM-TM2 (time mark).

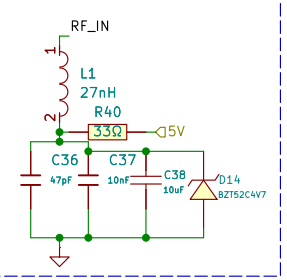
J7 grounding (1PPS from FE-5680A): J7 shell stays on GND. Power the FE-5680A from an isolated DC-DC converter, 16 V out, 40 W, fed from a battery (FE-5680A needs 15-18 V, 2 A during warm-up, 0.8 A steady). Its ground then reaches the board only through the J7 coax shield, so no rubidium supply current flows in the shield or across board GND. Mount the FE-5680A case isolated from the plate: insulating thermal pad, nylon screws. Do not use a non-isolated converter: its return would parallel the J7 shield and put rubidium supply current on board GND. GND is bonded to the box at the J2 arrestor. Battery negative must have no other connection to the box or plate. See docs/grounding-gps-sma.md.

IMPORTANT: The rubidium 1PPS clock input is optional and will be used in future experimentation. The unit will be placed inside the metal utility box along with the board, so no bulkhead connector. The FE-5680A runs off of 15-18V so will need a separate battery and DC-DC isolation.

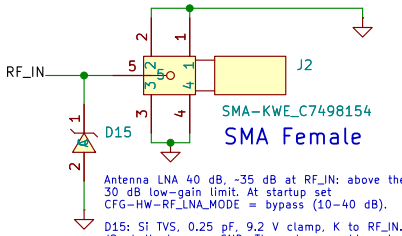
1PPS LED



RF INPUT



RF SMA Connector

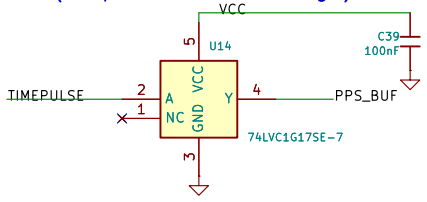


Antenna LNA 40 dB, ~35 dB at RF_IN: above the 30 dB low-gain limit. At startup set CFG-HW-RF_LNA_MODE = bypass (10-40 dB).

D15: SI TVS, 0.25 pF, 9.2 V clamp, K to RF_IN. J2 shell stays on GND. The antenna cable enters the utility box through a DC-pass coax arrestor bolted to bare metal on the box wall near the J2 corner, with a short coax jumper to J2. DC-pass is required: the antenna bias (5 V via R40/L1) must reach the antenna. The arrestor is the only GND-to-box bond. Earth the box there.

Antenna bias from 5V, not VCC_RF: the Waveshare L1/L2/L5 antenna needs 3.0-18 V at <=45 mA over 5 m RG174; VCC_RF (VCC-0.1 V, 50 mA max) cannot hold 3.0 V through any short-circuit resistor. R40 33R 1W limits a short to ~155 mA (L1 350 mA). C38 filters buck ripple. D14 4.7 V keeps RF_IN < 5.5 V DC abs max (antenna unplugged). Worst-case antenna voltage -3.3 V at 45 mA.

1PPS buffer (keeps SAFEBOOT_N high)



TIMEPULSE shares a pin with SAFEBOOT_N (1k inside U12): low at startup = safeboot, no NMEA (MAX-M10S IM 2.3.3.3: no load on it). R39/LED4 and R34 into GPIO16 (default pull-down) could hold it mid-rail. U14 leaves only input leakage on TIMEPULSE and drives the LED and GPIO16. VCC is from 3V3_CM4: no back-power path.

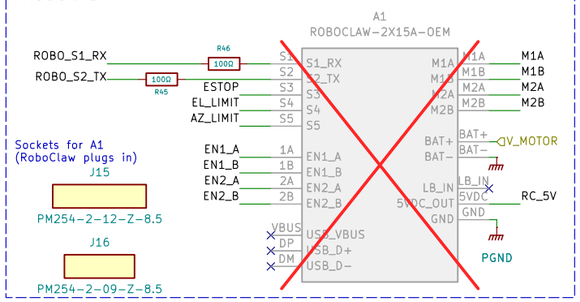
Sheet: /GPS/
File: gps.kicad_sch

Title: GPS 1PPS and RS232

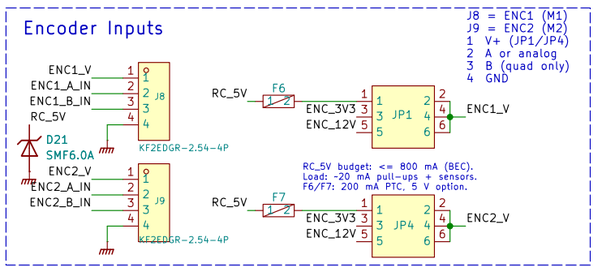
Size: A4	Date: 2026-08-18	**Rev: 2.0**
KiCad E.D.A. 10.0.1		Id: 7/13

ROBOCLAW-2X15A-OEM

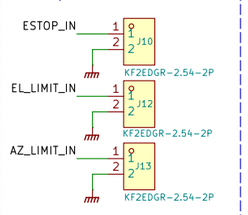
RoboClaw



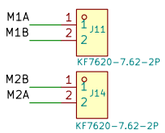
Encoder Inputs



E-stop / limit switches



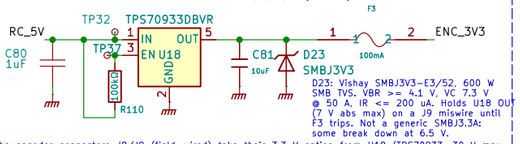
Motors Output
M1 = elevation
M2 = azimuth



J11 = Firgelli motor (elevation):
1 BLUE (+)
2 BROWN (−)
Swap to reverse, or
invert M1 in Motion Studio.

EMC: run motor leads in shielded cable, shield clamped 360 deg to the enclosure at the gland (not to J11/J14). Clip-on CM ferrite per pair at entry; 100 nF brush caps at the motors. Keep motor leads away from GNSS/encoder wiring.

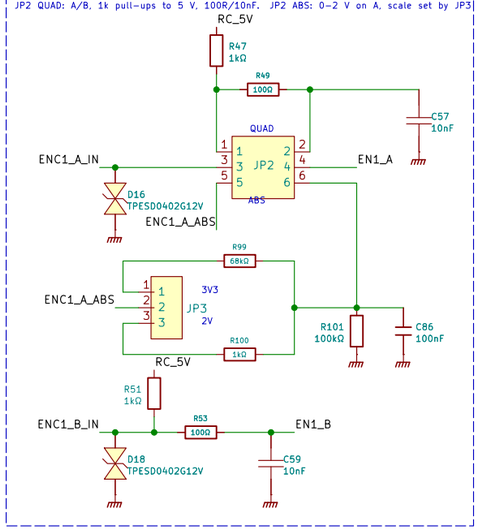
ENC_3V3: 3.3 V sensor supply (U18) for JP1 / JP4



The encoder connectors J8/J9 (field-wired) take their 3.3 V option from U18 (TPS70933, 30 V max, from RC_5V), not 3V3_CM4, so a miswired sensor supply cannot back-feed the CM4 or the GPS. U18 blocks reverse current, so it cannot reach RC_5V either. EN: R110 100k pull-up to RC_5V, TP37 for test (ground TP37 to hold U18 off). Never tie EN straight to IN.
F3 (100 mA PTC) sits between J9 and D23 (SMBJ3V3); F3 limits a shorted cable or a miswire onto J8/J9 while D23 clamps U18 OUT (7 V abs max). C81 >= 1.5 uF effective.
U18 EN (pin 3): R110 100k pull-up to RC_5V, TP37 for test. Ground TP37 to hold U18 off, release to run. EN clamps at 6.5 V: keep TP37 under 6 V and never tie EN straight to IN.

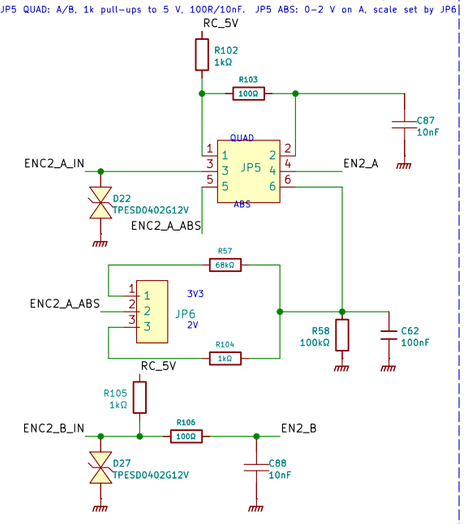
ENC1: A1 EN1 encoder input (motor M1)

JP2 QUAD: A/B, 1k pull-ups to 5 V, 100R/10nF. JP2 ABS: 0–2 V on A, scale set by JP3

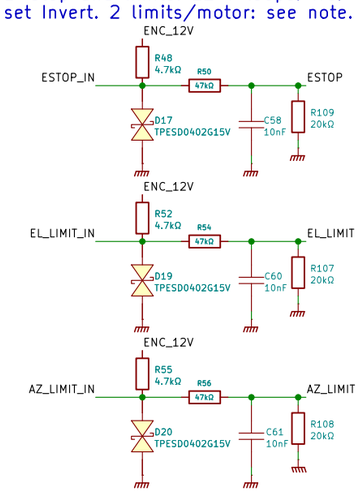


ENC2: A1 EN2 encoder input (motor M2)

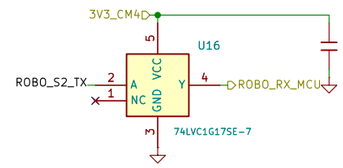
JP5 QUAD: A/B, 1k pull-ups to 5 V, 100R/10nF. JP5 ABS: 0–2 V on A, scale set by JP6



E-stop and limits: 12 V loops, NC, set Invert. 2 limits/motor: see note.

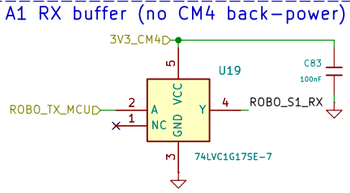


A1 TX buffer (no CM4 back-power)

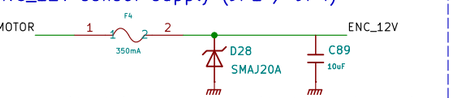


U16 isolates A1 S2 TX from CM4 GPIO13 (ROBO_RX_MCU); U19 (below the J9 box) isolates CM4 GPIO12 (ROBO_TX_MCU) from A1 S1 RX and its pull-up. A1 runs from the battery while the CM4 can be off (SW1 open, GLOBAL_EN power cycle). Wired direct, A1 would back-power the CM4 through GPIO13 or GPIO12 (abs max VREF+0.5 V). Both 74LVC1G17 run on 3V3_CM4; with it off, Ioff makes Y high-Z and A tolerates 3.3 V.

A1 RX buffer (no CM4 back-power)



ENC_12V sensor supply (JP1 / JP4)



Battery voltage (10–14.6 V) for sensors that need it (E6B2-CWZ6C: 5–24 V, 80 mA) and the E-stop / limit-switch loops (4.7k pull-ups). From V_MOTOR, so the return stays on the RoboClaw ground. F4 350 mA PTC, D28 SMAJ20A, C89 10 uF. Encoder pull-ups: RC_5V.

JUMPERS (ENC1: JP1 JP2 JP3, ENC2: JP4 JP5 JP6)

JP1/JP4 V+ supply 1–2: 5 V (RC_5V via F6/F7, 200 mA PTC)
3–4: 3.3 V (ENC_3V3, U18)
5–6: 12 V (ENC_12V, battery)
Shunt across a row only: 1–3 or 3–5 shorts two rails.
JP2/JP5 mode 1–3 + 2–4: QUAD
3–5 + 4–6: ABS
JP3/JP6 ABS scale 1–2: 3.3 V sensor -> 2 V
2–3: 0–2 V direct
Default ENC1 (Firgelli): 5V, QUAD
Default ENC2 (AS5600): 3.3V, ABS, 3V3

F6 / F7: 200 mA PTC (TECHFUSE SMD0603-020-24V, 0.55 ohm) in series with the 5 V option only (JP1 / JP4 1–2). RC_5V is the RoboClaw BEC, which also feeds the RoboClaw logic: a shorted or miswired encoder cable on the 5 V setting would drag it down. F6/F7 trip at 500 mA and reset when the fault clears. Hold current covers an E6B2-CWZ6C (80 mA) with margin: drop at 80 mA is about 50 mV. The 3.3 V option has F3 (100 mA), the 12 V option F4 (350 mA). The 1k pull-ups stay on RC_5V unfused (20 mA worst case).

Limit switches, two per motor

LIMIT SWITCHES: the RoboClaw has one limit input per motor (S4 = M1, S5 = M2) and supports two switches on it. Wire two NC switches in SERIES per axis (J12 elevation, J13 azimuth): either one opening lets the 4.7k pull-up raise the input. Set Invert (active-high). Both switches share the pin, so the RoboClaw takes fwd/rev from the commanded direction: set S4/S5 to Limit(Both), or to Home(User)/Limit(Fwd) with the retract-end switch on the reverse side so it also zeroes the encoder (elevation, Firgelli). NO switches would go in parallel instead. J12/J13 are 2-pin: daisy-chain the second switch in the field wiring.
The E-stop and limit loops run at 12 V (ENC_12V): R48/R52/R55 4.7k pull-ups pass about 2.6 mA through the closed contacts (contact wetting on long cable runs); open, the node sits at battery voltage (10–14.6 V) and the 47k/20k divider (R50/R109, R54/R107, R56/R108) delivers 3.0–4.4 V to S3/S4/S5 (5 V tolerant); C58/C60/C61 10 nF, tau 0.14 ms.
D17/D19/D20 are 15 V ESD parts for the 12 V nodes.

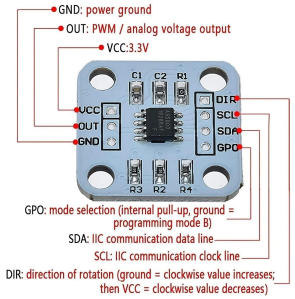
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 9/13

Notes About Motors + Axis Position Sensors

Azimuth Axis



Bringsmart JGY-370 Worm Gear Electric Motor 12V DC. Max 1.8A.



Typical azimuth encoder, AS5600

AZIMUTH SENSING: AS5600 analog absolute angle into ENC2 (A1 EN2, motor M2)

Jumpers: JP4 3–4 (V+ = 3.3 V), JP5 ABS (shunts 3–5 and 4–6), JP6 1–2 (3.3 V sensor –> 2 V).
Wiring, J9 to A1:
- J9–1 ENC2_V AS5600 supply: U18 3.3 V LDO (from RC_5V) via F3 (100 mA PTC), D23 clamp, JP4. Run the AS5600 at 3.3 V (VDD5V tied to VDD3V3 on the sensor board).
- J9–2 ENC2_A_IN AS5600 OUT. D22 ESD clamp –> JP5 –> JP6 –> R57 68k / R58 100k divider (x0.595) –> EN2_A (A1 2A). C62 100nF filters it (40.5k x 100nF, tau = 4 ms, about 39 Hz).
- J9–3 ENC2_B_IN Not used in ABS mode: leave open (R105 holds EN2_B high).
- J9–4 PGND Sensor return.

Scaling: AS5600 0–3.3 V –> 0–1.96 V at EN2_A, inside the RoboClaw 0–2 V absolute-encoder range.

How the RoboClaw uses it:
- Set EN2 to Absolute encoder mode in Motion Studio. The voltage is an absolute angle, so no homing is needed at power-up.
- M2 runs position PID on this reading. Speed is derived from the rate of change of position; there is no separate speed sensor.
- Read the RoboClaw position value at each end of travel and set those as the M2 min/max limits.
- The AS5600 is single-turn: its output jumps at the 0/360 deg wrap. Fit the magnet so the azimuth travel never crosses the wrap, or set ZPOS/MPOS (I2C or OTP) to place the range. A crossing makes the position loop run away.
- The magnet must be on a shaft that turns at most one revolution over the full azimuth travel.

AS5600 output loading (datasheet): full-range analog (the default) needs >= 100k. R57 + R58 = 168k, OK.
- OUT capacitive load 1 nF max: R57 isolates C62; keep the J9 cable short (about 100 pF/m).
- Divider source impedance is about 40k. If the RoboClaw EN2 pull-up stays on in Absolute mode it lifts the reading: bench-check EN2_A with and without the RoboClaw; calibrate limits on hardware.
- J9 power: U18 (off RC_5V) out, PGND return, all on the RoboClaw ground; nothing crosses NT1.

Elevation Axis



Firgelli F-SD-H-450-12V actuator. Max 5.5A. 450 lbs force.
https://www.firgelliauto.com/products/super-duty-actuators
There are no limit switches.
Feedback from Hall sensors.

Future addition? Firgelli makes a version with adjustable limit switches:
https://www.firgelliauto.com/products/adjustable-stroke-linear-actuators?variant=39421954261063
But only up to 220 lbs force.

ELEVATION SENSING: Firgelli Hall quadrature into ENC1 (A1 EN1, motor M1)

Jumpers: JP1 1–2 (V+ = 5 V), JP2 QUAD (shunts 1–3 and 2–4), JP3 not used.
Wiring, J8 to A1 (circuit: see ENCODER CHANNELS below):
- J8–1 ENC1_V RED Hall supply, 5 V from the RoboClaw BEC via F6 (200 mA PTC) and JP1 (RC_5V, shared). D21 SMF6.0A clamps RC_5V.
- J8–2 ENC1_A_IN YELLOW Hall 1. D16 ESD clamp –> JP2 –> R47 1k pull-up to RC_5V, R49 100R + C57 10nF –> EN1_A (A1 1A).
- J8–3 ENC1_B_IN WHITE Hall 2. D18 ESD clamp, R51 1k pull-up to RC_5V, R53 100R + C59 10nF –> EN1_B (A1 1B).
- J8–4 PGND BLACK Sensor return, on the RoboClaw ground.

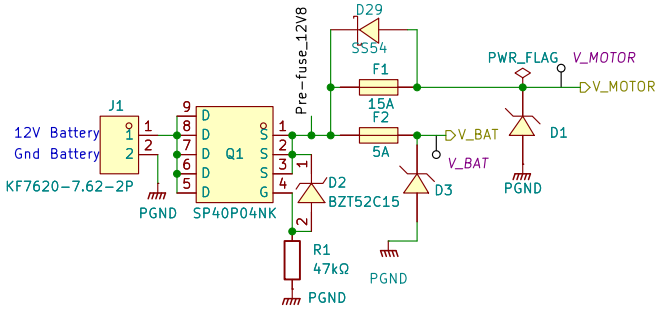
The pull-ups suit open-collector and push-pull Hall outputs. The 100R/10nF filters (tau = 1 us, about 160 kHz) remove cable noise and ringing. RoboClaw inputs tolerate 5 V: no level shifting.

How the RoboClaw uses it:
- The two Hall sensors are 90 degrees apart (quadrature A/B). The RoboClaw counts the pulses, and the A/B phase gives the direction. If the count falls while the actuator extends, swap A/B in Motion Studio; no rewiring is needed.
- Position = encoder count. Speed = count rate. Set QPPS (counts/s at full speed) in Motion Studio, then auto-tune the M1 velocity and position PID.
- The encoder is incremental, so the count is lost at power-off and the axis must be homed at start-up. The actuator has no limit switches. Home on the elevation limit switch (J12 –> S4, set as home/limit), or drive slowly to full retraction with a low current limit, then zero the counter.
- Stroke = counts / (Firgelli pulses per inch). Elevation angle follows from the mount geometry, which is non-linear, so convert between angle and counts on the CM4.

ENCODER CHANNELS ENC1 (J8) / ENC2 (J9): each accepts a quadrature or an absolute 0–2 V encoder, set per channel with shunts.
Connector: 1 V+ (JP1/JP4 1–2 = 5 V RC_5V via F6/F7 200 mA PTC, 3–4 = 3.3 V U18, 5–6 = 12 V battery) 2 A or analog in 3 B (quad only) 4 GND
Mode JP2/JP5 (2x3): QUAD = shunts 1–3 + 2–4: A goes through the 1k pull-up to 5 V and the 100R/10nF filter. ABS = shunts 3–5 + 4–6: A goes through JP3/JP6 into the 100k + 100nF node (JP3 1–2 = 68k top resistor, x0.595 for a 3.3 V sensor; 2–3 = 1k series, x0.99 for a source that is already 0–2 V, tau = 0.1 ms). B is always the quadrature path. In ABS mode the whole quadrature network, pull-up included, is lifted off the A input, so nothing pulls the analog node up.
Omron E6B2-CWZ6C (NPN open-collector, 5–24 V, 80 mA max, 35 mA sink, 100 kHz): JP1/JP4 = 12 V (5–6) or 5 V (1–2), mode QUAD. Wires: BROWN V+ –> 1, BLACK A –> 2, WHITE B –> 3, BLUE 0 V –> 4, ORANGE Z not used (insulate). Shield to the enclosure at the gland; it is not bonded to 0 V inside the encoder. The 1k pull-up (to 5 V, whatever V+ is) sinks 5 mA; the rising edge is tau = 11 us, fine to about 20 kHz: keep resolution x rpm / 60 under 20 kHz (1000 P/R to 1200 rpm). ENC_12V: V_MOTOR via F4 PTC, D28, C89.
Direct 0–2 V source (JP3/JP6 2–3): a 5 V potentiometer needs R1 = 1.5 x Rpot in series with its top end (RoboClaw manual); the 100k bottom resistor loads it by 1%.
RC_5V budget: the four 1k pull-ups draw up to 20 mA plus the sensors (Firgelli Hall ~10 mA, E6B2 80 mA, AS5600 7 mA); BEC limit 800 mA.

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12.8VDC Input Conditioning



Powered by a 12.8V 12Ah LiFePO4 Lithium battery IP67 Waterproof

V_MOTOR (A1 BAT+, via F1 15 A): motors 7.5 A max (F-SD-H-450 actuator 5.5 A + worm motor 2 A).
F2 5 A feeds the CM4 board branch (V_BAT) only.
F2/F1: Keystone 3557-2 holders; fit Littelfuse ATO 0287005.PXCN (5 A) / 0287015.PXCN (15 A).

V_BAT (J1): 12.8 V nom, 10.0 V min, 14.6 V max (charging).
Survives 18 V max (TVS). Reverse battery: Q1 P-FET blocks it; survives −14.6 V, nothing conducts, no fuse blows.
Q1 – SP40P04NK, 40 V P-channel MOSFET. Rated 85 A continuous at Tc 25 C (57 A at Tc 100 C), 340 A pulsed; Rds(on) 6.1 mOhm max at −10 V. Carries 8.7 A typ / 9.4 A worst: ~0.4 W typ, 0.55 W worst, ~0.8 W hot.
Give Q1's drain tab a solid copper pour.

F1/F2 are the backup for a reversed battery: they blow only if Q1 has failed short (about 270 A then flows through D1/D3 and the RoboClaw diodes). Check polarity and Q1 before replacing a blown fuse.

Standby drain: SW1 (power sheet) switches the CM4 / logic supply only. With SW1 off the battery still feeds the RoboClaw logic (~30 mA, via F1) and U11 + dividers (~1.3 mA): about 31 mA, so 12 Ah lasts ~16 days. For storage or transport unplug J1 (or remove F1/F2); charge first.

D29 (SS54, 5 A 40 V Schottky, top side between F1 and J14) sits across F1, anode on V_MOTOR: the return path for RoboClaw regenerative current when F1 is open (Basicmicro data sheet p.8). A fault that blows F1 pulls V_MOTOR low, so D29 cannot bypass the fuse. If the battery BMS blocks charging, set the RoboClaw max main battery voltage to 16 V so it stops regenerating before D1 clamps.

JRSeti		
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File: battery-input.kicad_sch		
Title: Battery Input		
Size: A4	Date: 2026-10-17	**Rev: 2.0**
KiCad E.D.A. 10.0.1		Id: 17/13